Evaluation Cubed: Judges Have Resolution Limits, and You Can Measure Them Without Labels

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Abstract

Language models are ranked by LLM judges, and judges are validated by meta-evaluation: agreement with gold labels on a pooled item set. Nobody evaluates the meta-evaluation. We build the third level—*Eval³*—and use it to ask whether meta-evaluation delivers what it is used for, namely trustworthy system rankings. Our benchmark sidesteps the annotation bottleneck that makes such studies impractical: answers are assembled from atomic factual claims, so true quality is known *by construction*, and a quality-preserving restyling gives label-free access to judge error. Across 28 judge configurations and 40,120 judgements we find, first, a negative result that reframes the problem: when systems differ by a large margin every judge ranks them correctly, so judges are not simply bad. What varies, enormously, is each judge's *resolution limit*—the smallest true quality gap it can order reliably once systems differ in surface form. Limits span $60\times$, from 0.8% to 50% of an answer's factual content. Meta-evaluation accuracy tracks this quantity ordinally ($\rho = -0.89$) but compresses a $60\times$ difference into 22 accuracy points, and cannot be converted into a limit: its residual error is 6.3% of factual content, larger than the entire limit of the seven best judges. We give a label-free estimator that can, with residual error 3.3% ($R^2 = 0.93$), and a per-comparison *certificate*: in the leaderboard regime it raises ranking accuracy from 0.860 to 0.969 while declining to certify the pairs on which accuracy is 0.721, using no human labels. We argue this is where the evaluation regress terminates: rankings depend only on quality *differences*, and transformations with known quality effect supply differences without supplying labels.

1 Introduction

Almost every claim about which language model is better now passes through an LLM judge. Because judges are known to be fallible, the field validates them by *meta-evaluation*: collect human preference labels, ask how often the judge agrees, and report that agreement [Zheng et al., 2023, Lambert et al., 2024, Tan et al., 2025]. Meta-evaluation is itself an evaluation procedure, with its own design choices—which items, which systems, which statistic—and those choices are not themselves evaluated against anything.

This is uncomfortable for a reason that is easy to state and hard to dismiss. Meta-evaluation scores a judge by its *agreement with labels*. Practitioners use a judge to *rank systems*. These are different quantities, and the field has recently accumulated evidence that the link between them is loose: judge rankings shift by up to 14 positions across meta-benchmarks [Norman et al., 2026], exact-match agreement systematically overstates discrimination [Rao and Callison-Burch, 2026], and judges respond to cosmetic rewrites about as strongly as to meaningful ones [Weng et al., 2026]. What is missing is an account of *what* the loose link costs a practitioner, and a way to bound that cost.

Preprint. ∗ indicates equal contribution.

We supply the missing level. Writing L1 for evaluating a system, L2 for evaluating a judge, this paper is L3: evaluating the meta-evaluation, judged against the decision it actually informs. We call it *Eval³*.

The obstacle, and how we get around it. Any L2 or L3 study needs ground truth, and collecting it by annotation is exactly what makes such studies rare. We avoid annotation entirely by *constructing* it. Each item is a question plus six atomic, independently checkable factual claims. A system is defined by how many of those claims are replaced by minimally corrupted variants (a wrong date, a swapped name, a reversed relation), so its true quality is a design parameter rather than an estimate. Orthogonally, the same claim set is rendered in three styles. Since the content is held fixed, restyling is quality-preserving *by construction*, and any judge score difference across styles is judge error—measurable with no labels at all. This design also has the property that its faults are mechanically catchable, which recent work argues is the decisive advantage of perturbation-based corpora over LLM-generated ones [Ballı, 2026]; we run their integrity protocol and report it, including a fault it caught (Section 3.1).

What we found, including what we expected and did not find. Our first result is negative and it reframes everything after it. At the quality gaps our grid was originally built to test—up to four of six claims—nearly every judge ranks systems perfectly (system ranking accuracy 1.000 for 18 of 24 pointwise configurations; mean 0.977). Judges are not broadly incompetent at ranking. We had also expected that pooled agreement would fail to predict ranking validity, and that restricting a meta-benchmark to clear-cut, stylistically matched pairs would destroy the signal. Neither held: agreement predicts ordinally ($\rho = -0.89$) and does so robustly across every composition we ablated (ρ from -0.79 to -0.89). We report these because they constrain the claim that survives.

What survives is sharper. Every judge has a **resolution limit**: a true quality gap below which its ranking can be reversed by nothing but surface form. These limits span $60\times$ across our judge pool, so the same 5% leaderboard gap is decisive under one judge and meaningless under another—and current practice reports no quantity from which you could tell which. Meta-evaluation accuracy is ordinally informative but not *calibrated*: mapping it to a limit leaves a residual of 6.3% of factual content, exceeding the entire limit of the seven best judges we tested.

Contributions.

1. **Eval³**, a benchmark and harness for evaluating meta-evaluation *protocols*, built on construction-based ground truth so that no human annotation is required at any level (Section 3).
2. A decomposition showing that only *differential* judge bias can corrupt a ranking, and a proof that no pooled meta-evaluation metric can determine ranking validity, because such metrics are invariant to a group that ranking fidelity is not invariant to (Section 2).
3. The **resolution limit** as the decision-relevant summary of a judge, its measurement, and the empirical finding that it varies $60\times$ while accuracy varies 22 points (Section 4).
4. A **label-free estimator** of the limit ($R^2 = 0.93$, residual 3.3% versus 6.3% for the best gold-label predictor) and a per-comparison **certificate** that raises ranking accuracy from 0.860 to 0.969 at 55.8% coverage (Sections 2, 6).
5. An argument that the evaluation regress *terminates* at L3, because the external grounding L3 requires is $O(1)$ in the number of systems, judges and items (Section 8).

2 Setup and theory

Let systems be s drawn from a population $\mathcal{P}$ and items i from a distribution $\mathcal{I}$. Write $q(s,i) \in [0,1]$ for true per-item quality and $Q(s) = \mathbb{E}_i q(s,i)$ for true system quality. A judge J returns $\hat{q}(s,i)$, giving $\hat{Q}(s) = \mathbb{E}_i \hat{q}(s,i)$. Define judge error $e(s,i) = \hat{q}(s,i) - q(s,i)$ and its item-average $\bar{e}(s) = \mathbb{E}_i e(s,i)$, and split

$$\bar{e}(s) = \mu + \delta(s), \qquad \mu = \mathbb{E}_s \bar{e}(s), \quad \textstyle\sum_s \delta(s) = 0. \tag{1}$$

Proposition 1 (Only differential bias can corrupt a ranking)**.** *For any two systems,* $\hat{Q}(s) - \hat{Q}(s') = \big(Q(s) - Q(s')\big) + \big(\delta(s) - \delta(s')\big)$. *The global bias* μ *cancels exactly, and the idiosyncratic residual*

$e(s,i) - \bar{e}(s)$ has zero mean and affects only the sampling variance of $\hat{Q}$, not its expectation. Hence the judge orders the pair correctly whenever $|\delta(s) - \delta(s')| < |Q(s) - Q(s')|$.

Judge error therefore splits into three parts with entirely different consequences: μ is harmless for ranking, the idiosyncratic part costs statistical power but not correctness, and δ is the whole problem. Meta-evaluation accuracy is a monotone function of $\mathbb{E}|e|$, which mixes all three. This is the source of the difficulty, and it is not repaired by choosing a better agreement statistic.

2.1 Pooled meta-evaluation cannot determine ranking validity

Definition 2 (Pooled metric)**.** A meta-evaluation metric is *pooled* if it is a function only of the multiset $\{(\hat{q}(x), q(x)) : x \in \mathcal{X}\}$ of (judge score, gold label) pairs over the evaluation set, discarding which system produced each x.

Item-level agreement, pairwise preference accuracy, correlation with gold, Cohen's κ and Brier score are all pooled, as is essentially every number reported by current meta-benchmarks.

Theorem 3 (Pooling destroys the relevant information)**.** *Let G be the group of permutations that exchange the judge scores of two cells carrying the same gold label, possibly belonging to different systems. Every pooled metric is G-invariant, whereas $\hat{Q}(\cdot)$, and hence the induced ranking, is not. Consequently, if $\mathcal{P}$ contains two systems sharing a gold-label value on at least m items each, and the judge's scores are non-constant within that label class, then for every pooled metric M there exist judges J_1, J_2 with $M(J_1) = M(J_2)$ but $\mathrm{SRA}(J_1, \mathcal{P}) = 1$ and $\mathrm{SRA}(J_2, \mathcal{P}) = 0$, provided the within-label score spread scaled by m/n exceeds the true quality gap.*

Proof sketch. Applying an element of G leaves the multiset of (score, label) pairs unchanged, so M is unchanged. Each swap that moves a high judge score from the better system's label-c cells to the worse system's label-c cells decreases $\hat{Q}(\text{better}) - \hat{Q}(\text{worse})$ by a fixed amount. Accumulate swaps until the sign flips. Full proof in Appendix A. □

The reading is not that current meta-benchmarks are carelessly designed. It is that no amount of care *within the pooled frame* can settle the question they are used to settle. This is consistent with, and explains, the empirical instability reported by Norman et al. [2026], and it is why we look for a quantity outside that frame.

2.2 Resolution limits and a label-free certificate

Definition 4 (Quality-preserving family)**.** A transformation T is *quality-preserving* if $q(T(x)) = q(x)$ for every output x. Given a family $\mathcal{T}$ of such transformations, the judge's *style range* on $\mathcal{P}$ is $R_J = \max_{T,T' \in \mathcal{T}} \big| \mathbb{E}_s \hat{Q}(T \circ s) - \mathbb{E}_s \hat{Q}(T' \circ s) \big|$.

R_J requires no quality labels. It uses only the fact that $\mathcal{T}$ preserves quality, which is a property of transformations we wrote, not an annotation of any system under test.

Theorem 5 (Perturbation certificate)**.** *Suppose the differential bias is generated by the family, i.e. $\max_s \delta - \min_s \delta \le R_J$. Then for any pair with $|\hat{Q}(s) - \hat{Q}(s')| > R_J$ the judge's ordering is correct.*

Proof. Say $\hat{Q}(s) > \hat{Q}(s')$. By Proposition 1, $Q(s) - Q(s') = (\hat{Q}(s) - \hat{Q}(s')) - (\delta(s) - \delta(s')) > R_J - R_J = 0$. □

Dividing R_J by the judge's sensitivity to real quality converts it into the units of the quality construct.

Definition 6 (Resolution limit)**.** Let β_J be the judge's score change per unit of true quality, estimated by applying a degradation of *known* magnitude. The judge's resolution limit is $\delta_{\text{style}}(J) = R_J/\beta_J$: the true quality difference that the judge's preference over surface form can counterfeit.

Both R_J and β_J are estimable without quality labels—the first from restyling, the second from applying a known corruption—so δ_{style} is label-free while being expressed in interpretable units ("fraction of an answer's factual content").

Scope, stated plainly. Theorem 5 is sound *relative to* $\mathcal{T}$. It bounds the component of δ generated by the perturbations actually tested and is silent about a confound orthogonal to them. We regard this as the honest form of the guarantee rather than a weakness: it replaces an unfalsifiable "trust the judge" with an explicit, extensible list of confounds, and widening $\mathcal{T}$ strengthens the guarantee monotonically. Section 6 measures the residual empirically.

3 The Eval^3 benchmark

Items. We generated 96 questions across 32 topics, each with six atomic claims and a minimally corrupted variant of each claim, labelled by corruption type (quantity, date, entity, relation, mechanism) with at least three types per item. An independent verification pass checked that every true claim is true and every corrupted claim false; items where any of the six pairs failed were dropped, leaving **85 items and 510 verified claim pairs**. Mean claim length is 12.7 words.

Systems. A system is a cell of a balanced 5×3 design: quality level $k \in \{0, \dots, 4\}$ (how many of the six claims are corrupted; nested, so quality is monotone in k within every item) crossed with style $\in$ {plain, polished, padded}. Content is fixed by (item, k) and shared across styles. True quality is $(6-k)/6$ by construction. Rendering is done by a single fixed model so that rendering skill cannot vary across cells, giving $85 \times 5 \times 3 = 1275$ answers.

Continuous quality gaps at zero additional cost. A system need not corrupt every item equally. Defining a system by its *mean* corruption level m—a fraction of items at $k = \lfloor m \rfloor$, the rest at $\lceil m \rceil$—yields any true quality gap we like, with a judge score averaged over cells already collected. The continuous quality axis therefore costs no additional API calls.

Judges. Eight models spanning a wide capability range (`gpt-4.1-nano`, `gpt-4o-mini`, `gpt-4.1-mini`, `gpt-4.1`, `gpt-5-nano`, `gpt-5-mini`, `gpt-5.4-nano`, `gpt-5.4-mini`) $\times$ three pointwise protocols (DIRECT 1–10, RUBRIC, COT), plus a pairwise-against-anchor protocol on four models run in both presentation orders. That is **28 judge configurations** and **40,120 judgements** (30,600 pointwise, 9,520 pairwise). Unparseable rate: 0 pointwise, 0.75% pairwise.

3.1 Stimulus integrity, and a fault it caught

Ballı [2026] show that a silently failing generation step can manufacture an effect that does not exist ("disappearance") or bend the magnitude and even the direction of one that does ("distortion"), and that aggregate statistics cannot distinguish the two. We ran their protocol before computing any statistic.

Mechanical no-op check: $0/510$ corrupted claims string-identical to their true counterpart. Word counts: flat across quality levels within each style (max deviation 2%). Degeneration: no fragments under three words anywhere.

The check earned its keep on the fourth item. In the first build, **61–68 of 85 `plain` answers were verbatim concatenations of the claim list**: the instruction ("4–7 short sentences, no formatting") was satisfiable by emitting the six claims one per sentence. That condition was not a rewrite but the raw stimulus, and comparing a bare fact list against prose would have inflated our central quantity—precisely the distortion mode Ballı [2026] describe. We forbade one-claim-per-sentence restatement and re-rendered: 0 verbatim copies in all 15 conditions, and `plain` rose from 77 to 99 words. Final claim fidelity is $7621/7650 = 99.6\%$, with verified true-claim counts of $6.00/5.02/4.01/3.01/1.99$ against intended $6/5/4/3/2$. All statistics below use the corrected corpus; we also read 20 raw items per condition by hand (Appendix B).

4 Judges are not bad at ranking; they have resolution limits

Observation 1: at large gaps, ranking is essentially solved. On the original grid, where systems differ by one to four of six claims, mean system ranking accuracy across the 15-system grid is 0.977, and is exactly 1.000 for 18 of the 24 pointwise configurations. A two-way ANOVA on each judge's system means decomposes its between-system variance into a quality component (valid signal) and

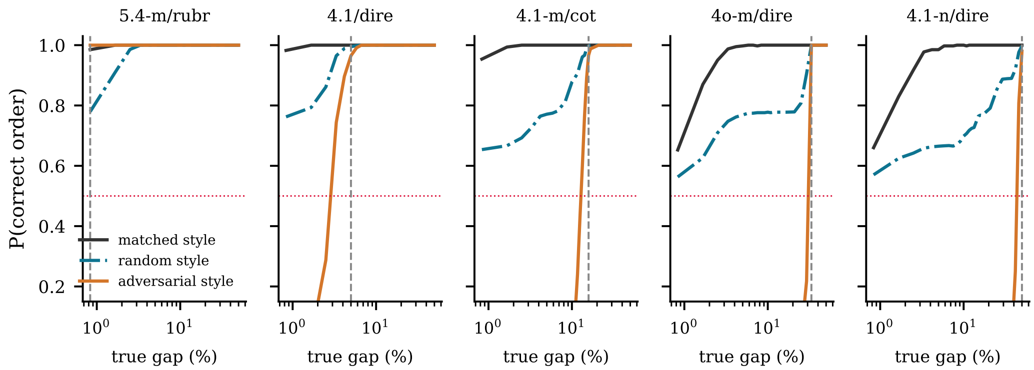

Figure 1: Probability of correctly ordering two systems as a function of their true quality gap (log axis), for five representative judges ordered by resolution limit. Black: both systems rendered in the same style. Teal: styles assigned independently at random. Orange: the worse system wears the judge's preferred style. Dashed grey marks the measured resolution limit. **Takeaway:** the adversarial curve is a step, and its location—not the judge's average accuracy—is what determines whether a leaderboard gap means anything.

style plus interaction components (pure bias, since restyling preserves quality by construction). The quality share ranges from 0.684 to 0.998 (Figure 5). *Interpretation:* quality dominates surface form for every judge we tested, and with 85 items the law of large numbers does the rest. Claims that LLM judges cannot rank systems are, in this regime, not supported.

Observation 2: the picture inverts at small gaps. Figure 1 sweeps the true quality gap continuously and plots the probability that a judge orders a pair correctly, under three style regimes: both systems in the same style, styles assigned independently at random, and an adversarial assignment in which the *worse* system wears the judge's preferred style. Under matched styles, accuracy climbs quickly. Under adversarial assignment it is pinned near zero until the true gap exceeds a threshold, then steps to one—exactly the step Proposition 1 predicts, since a fixed style pair contributes a constant offset. The random-assignment curve, the realistic case for a heterogeneous leaderboard, sits between the two and approaches one only slowly.

Observation 3: limits vary by $60\times$; accuracy varies by 22 points. We define a judge's measured resolution limit as the smallest gap beyond which adversarial accuracy stays above 0.95 for all larger gaps. Figure 2(a) reports it for all 28 configurations. The best, `gpt-5.4-mini`/RUBRIC, resolves 0.83% of an answer's factual content; the worst, `gpt-4.1-nano`/DIRECT, needs 50%. The median is 10.8%. Over the same pool, pooled meta-evaluation accuracy spans 66.1% to 88.3%.

Interpretation. Accuracy and the limit are strongly related in rank order ($\rho = -0.89$), so accuracy is not uninformative—a hypothesis we entered with and abandoned. But the accuracy scale is severely non-linear in the decision-relevant quantity: a 22-point accuracy range corresponds to a $60\times$ range in resolving power (Figure 2(b)). Reading "this judge scores 0.69 and that one 0.88" as a modest difference is a serious misreading of what those numbers imply for a leaderboard.

Observation 4: verbosity preference is the norm, and it reverses at the frontier. Of 28 configurations, 18 score the `padded` style highest, 7 `plain` and 3 `polished`, with the exceptions concentrated in the newest family. Protocol matters as much as model: within `gpt-5.4-mini` the limit is 0.83% under RUBRIC, 2.5% under DIRECT, 5.0% under COT and 10.8% under pairwise. *Interpretation, offered cautiously:* our pairwise arm shows *larger* style sensitivity than pointwise scoring, which sits awkwardly beside Norman et al. [2026]'s small (< 0.011) verbosity bias under a pairwise rubric. Our pairwise score is a win rate against a fixed anchor—a different estimator on a different scale—so we flag this as needing direct study rather than treating it as a contradiction.

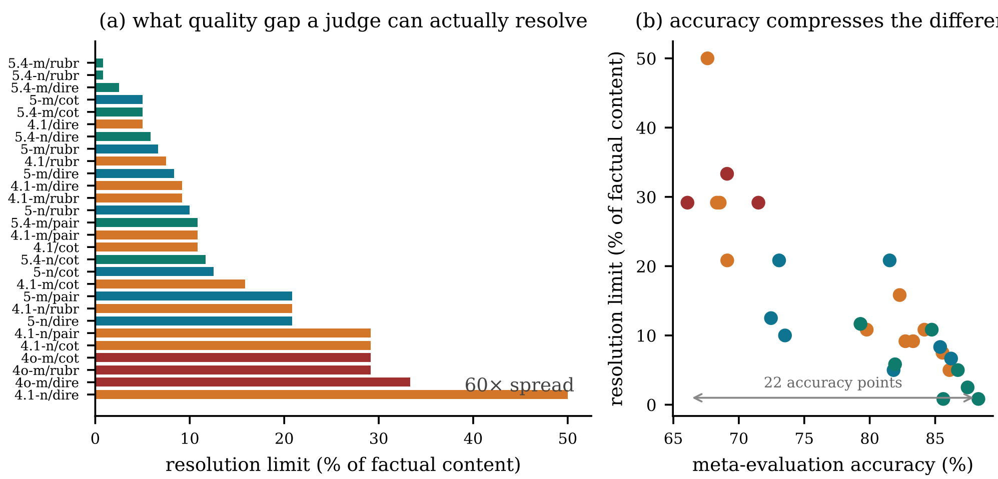

Figure 2: **(a)** Measured resolution limit for all 28 judge configurations, coloured by model family. **(b)** The same limits against pooled meta-evaluation accuracy. **Takeaway:** the quantity that decides whether a leaderboard gap is meaningful varies by $60\times$ across judges, and the accuracy scale compresses that into 22 points.

Table 1: Predicting a judge's resolution limit. Residual standard deviation is in units of true quality (fraction of an answer's factual content); lower is better. Gold-label predictors require an annotated meta-benchmark; δ_{style} requires none.

Predictor	Labels needed	R^2	Residual s.d. (% of content)
Accuracy, full composition	gold	0.723	6.3
Accuracy, clear-cut pairs	gold	0.699	6.6
Accuracy, style-matched pairs	gold	0.684	6.7
Accuracy, clear-cut + matched	gold	0.667	6.9
Accuracy, near-ties only	gold	0.698	6.6
δ_{style} (ours)	**none**	**0.927**	**3.3**

5 What the meta-benchmark can and cannot tell you

We expected the informativeness of pooled agreement to depend on how the meta-benchmark is composed, and specifically that restricting it to clear-cut pairs—the standard design goal, since ambiguous pairs are unreliable to annotate—would destroy the signal. **It does not.** We recomputed accuracy under five compositions and correlated each with the resolution limit: *full* $\rho = -0.89$, *clear-cut only* ($|\Delta k| \geq 3$) $\rho = -0.80$, *style-matched pairs only* $\rho = -0.87$, *both restrictions* $\rho = -0.79$, *near-ties only* ($|\Delta k| = 1$) $\rho = -0.89$. The ordinal signal is robust.

The failure is not ordinal but *metric*. Table 1 regresses the resolution limit on each predictor and reports the residual spread in the units that matter.

Interpretation. A residual of 6.3% of factual content is not a small error. It exceeds the entire resolution limit of the six best judges in our pool. So a practitioner who knows a judge's meta-benchmark accuracy still cannot answer the question they need answered: *is my observed* 5% *leaderboard gap real?* The label-free estimator halves that residual and is expressed directly in the units of the question (Figure 3).

6 The certificate in the leaderboard regime

The regime that matters is the one leaderboards live in: systems separated by small margins, each rendered in whatever style it happens to produce. We drew 4,000 random system pairs per judge with

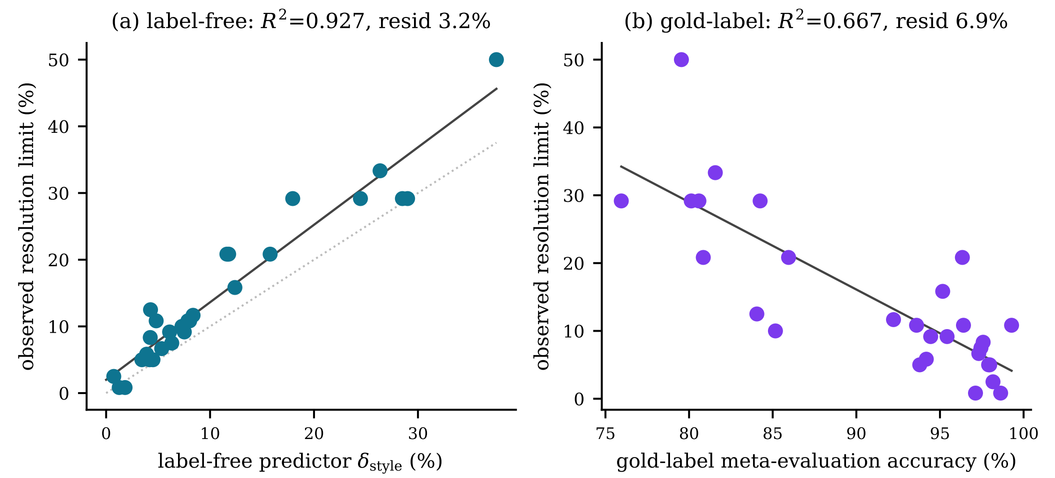

Figure 3: Recovering the resolution limit. **(a)** The label-free predictor δ_{style}; dotted line is $y = x$. **(b)** Gold-label accuracy on a clear-cut, style-matched meta-benchmark. **Takeaway:** both are informative, but only the label-free predictor is calibrated well enough to convert into a decision about a specific leaderboard gap.

Table 2: Certificate performance, averaged over 28 judge configurations, $4{,}000$ pairs each. The threshold R_J is estimated from data, so the right-hand block re-estimates it on 42 items and evaluates on the disjoint 43. That split also halves the evaluation set, which independently adds sampling noise to every system mean; the two effects are reported together rather than disentangled.

	all 85 items		split-half
	Mean	Worst judge	Mean
Ranking accuracy if you simply trust the judge	0.860	0.687	0.838
Coverage (fraction of pairs certified)	0.558	—	0.571
Ranking accuracy on *certified* pairs	**0.969**	0.834	**0.943**
Ranking accuracy on pairs it *declines*	0.721	—	0.683

true quality gaps uniform in $(0.8\%, 16.7\%]$ of factual content and styles drawn independently, then applied Theorem 5: certify the ordering iff $|\hat{Q}(s) - \hat{Q}(s')| > R_J$, with R_J measured with no labels.

Observation. The certificate separates the trustworthy claims from the rest: accuracy is 0.969 on the pairs it certifies and 0.721 on the pairs it refuses, a gap of 24.8 points, at 55.8% coverage and zero labelling cost. The separation is not an artefact of estimating the threshold on the evaluation data: with R_J estimated on a disjoint half of the items, certified accuracy is 0.943 against 0.683 on the declined pairs, a gap of 26.0 points. For the strongest judge (`gpt-5.4-mini`/DIRECT) coverage reaches 97.2% at 98.8% accuracy; for the weakest it correctly collapses to 12.5% coverage, which is the right behaviour—a judge that cannot resolve the gaps in question should certify almost nothing.

Interpretation, including the shortfall. Theorem 5 would give 100% certified accuracy if δ were generated entirely by the style family. The observed 96.9% therefore *measures* the residual differential bias $\delta_\perp$ orthogonal to our three transformations. That 3.1% is the price of a finite $\mathcal{T}$, and it is the quantity a practitioner reduces by adding transformations. We prefer reporting it to asserting a guarantee we have not earned.

7 Real systems: the regime leaderboards actually occupy

Constructed systems buy exact ground truth; the obvious objection is that they are not real systems. We repeat the study with five models answering the same 85 questions freely, whose quality is *measured*—reference claims supported minus contradicted—rather than designed, then restyle each real answer into the same three styles so R_J stays computable without labels.

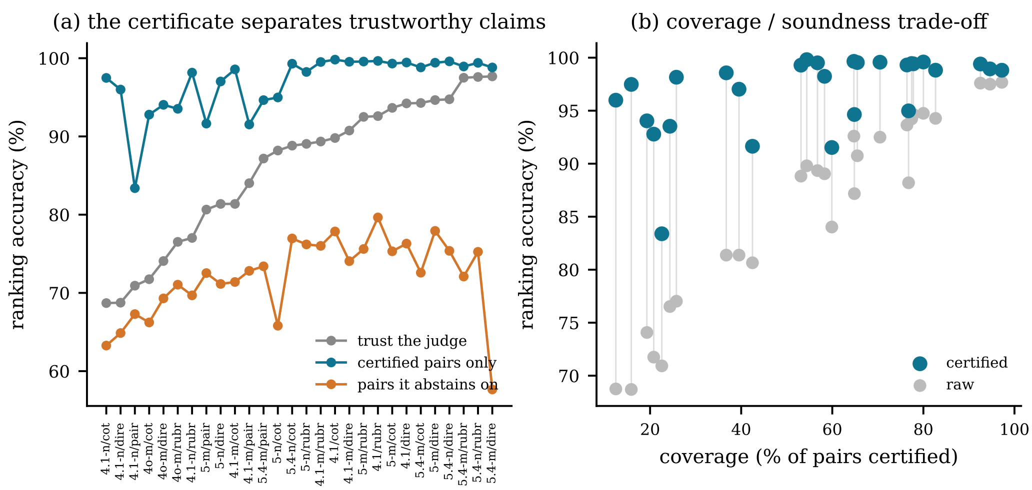

Figure 4: The certificate in the leaderboard regime (gaps $\leq 16.7\%$ of factual content, independent styles). **(a)** Judges sorted by raw accuracy. **(b)** Coverage against accuracy. **Takeaway:** certified orderings are reliable across the whole judge pool, the pairs the certificate declines are exactly the unreliable ones, and weak judges correctly certify almost nothing.

Two caveats bound what this shows. Measured quality is a coverage-weighted proxy (correlating with length, $r = 0.31$ item-level), and restyling a real answer moves it by 0.049, so quality-preservation is approximate rather than exact. Both push these numbers toward noise, which is why the constructed arm carries the main claims.

Observation. The five models span a measured quality range of $\mathbf{0.108}$—almost exactly the median judge's resolution limit (0.108). Real comparisons sit right at the edge of what these judges resolve, and the consequences show: judges agree with measured quality at $\rho = 0.85$ but with each other at only $\rho = 0.70$, and raw ranking accuracy on fine-grained mixture pairs falls to 0.652. The certificate still separates, 0.754 certified against 0.584 declined, at 36.0% coverage.

Interpretation. This is what the theory predicts for a population separated by less than the resolution limit, and it is the ordinary case, not an adversarial one. We do not read 0.754 as validating the certificate at the level the constructed arm supports; with a noisy proxy and approximate quality-preservation the ceiling here is well below one. What the section establishes is that the regime the constructed arm studies is the regime real comparisons occupy.

8 Where the regress terminates

The worry motivating this paper is a regress: judges need meta-evaluation, meta-evaluation needs meta-meta-evaluation, without ground. It terminates at L3, and Proposition 1 says why. **Rankings are invariant to the level of quality; they depend only on differences.** Levels require external grounding; differences do not, because a transformation with a *known* effect on quality supplies signed information about differences without ever naming a quality value.

The precise claim is about how grounding scales. L1 needs labels for every item, $O(\text{items})$. L2 needs a labelled meta-benchmark, also $O(\text{items})$, re-collected whenever the system population changes. L3 needs only that $\mathcal{T}$ is quality-preserving—a fixed, reusable assumption about transformations we authored, $O(1)$ in systems, judges and items. A level-4 question ("is $\mathcal{T}$ really quality-preserving?") is a single finite check, not a demand that grows with what is evaluated. The regress terminates not because we run out of patience but because the grounding requirement stops growing.

9 Related work

Meta-evaluation of judges. LLM-as-judge and its biases were charted by Zheng et al. [2023]; benchmarks followed [Lambert et al., 2024, Tan et al., 2025, Zeng et al., 2024], along with surveys [gu2, 2024] and large cross-task studies [bav, 2024]. The closest empirical neighbour is Norman et al. [2026]: 21 judges, $\sim$541k judgements, showing universal κ deflation, judge rankings shifting by up to 14 positions across meta-benchmarks, and a consistency–bias paradox. They establish that judge rankings are unstable; we ask what that costs at the level of a system ranking, explain it via Theorem 3, and give a label-free remedy. Rao and Callison-Burch [2026] show that five agreement statistics collapse to one value on binary verdicts; their scope is single-judge validation, not whether agreement predicts system ranking, and Table 1 supplies that link. poo [2026] independently argue that pooled reporting hides system-specific structure.

Bounding judge bias. Feuer et al. [2026] propose average bias-boundedness, an algorithmic framework that *enforces* a guarantee by modifying the evaluation, validated on Arena-Hard-Auto. Ours is a *post-hoc certificate* on an ordering already produced, requiring no change to the judge; the two compose. Weng et al. [2026] are the closest neighbour to our label-free diagnostic: a Policy Invariance Score from certified-equivalent rubric rewrites. Three differences matter. Their rewrites are certified by three human annotators, ours are quality-preserving by construction; theirs are rubric-side, ours output-side; and, in their own words, they do not bound downstream ranking error—exactly the gap Theorem 5 fills. Perturbation and metamorphic stress-testing [jud, 2026, per, 2024, sin, 2025] is the ancestry of our transformation family.

Benchmark validity. A systematic review of 445 benchmark papers found only 16% report uncertainty estimates [con, 2025]; we bootstrap every headline number over items. Validity audits of specific benchmark families [too, 2026] and older work on the validity of automatic metrics [Reiter, 2018, Chaganty et al., 2018] and on measurement modelling [Jacobs and Wallach, 2021] frame the construct question our declared transformation family operationalises.

10 Limitations

One provider. All judges and all generation come from a single model family; no Anthropic API key was available in our environment. Our claims concern the *protocol* rather than any vendor, and the pool spans a wide capability range, but we make no cross-provider claim and replication there is the obvious next step.

A narrow quality construct. True quality here is the fraction of six atomic factual claims that are correct. This is deliberate—it is the dimension along which ground truth is *constructible*—but our results speak to ranking on factual accuracy, not helpfulness, style, or safety.

Is restyling really quality-preserving? A reader may hold that verbosity genuinely degrades an answer, making `padded` not quality-preserving. We re-ran everything with the length-matched family $\{$plain, polished$\}$: limits still span $35\times$ and the estimator still recovers them ($R^2 = 0.876$, residual 2.5%; Appendix D). More fundamentally, the framework *requires* declaring $\mathcal{T}$, forcing an explicit commitment to a quality construct rather than leaving it implicit. The certificate's 3.1% shortfall from soundness measures confounds outside those transformations, reported not hidden.

The real-system arm is weaker than the constructed one. Section 7 rests on a coverage-weighted quality proxy and only approximate quality-preservation. Read it as showing *which regime* real comparisons occupy, not as a second validation. A stronger test needs an independently defensible quality construct—the same problem that motivated construction-based ground truth in the first place.

11 Conclusion

Meta-evaluation asks how often a judge agrees with a label. Practitioners need to know whether a leaderboard gap is real. We showed these come apart in a specific, measurable way: judges have resolution limits varying $60\times$ across a single provider's model range, agreement tracks those limits ordinally but compresses them into 22 accuracy points and cannot be converted into one, and a label-free perturbation estimator both predicts the limit ($R^2 = 0.93$) and certifies individual ranking claims ($0.860 \rightarrow 0.969$ accuracy at 55.8% coverage). The practical recommendation is small and concrete: alongside judge accuracy, report the judge's resolution limit in the units of your quality construct, and do not defend a leaderboard gap smaller than it.

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A Proof of Theorem 3

Fix a population containing systems s_1 (the better) and s_2, each evaluated on n items. Suppose both have at least m items carrying the same gold-label value c, and let H_1, H_2 be those label-c cell sets, $|H_1|, |H_2| \geq m$.

Let σ be the transposition that exchanges the judge scores of one cell $x_1 \in H_1$ and one cell $x_2 \in H_2$, leaving all gold labels in place. Because $q(x_1) = q(x_2) = c$, the multiset $\{(\hat{q}(x), q(x))\}$ over the whole evaluation set is unchanged by σ: the pair $(\hat{q}(x_1), c)$ and the pair $(\hat{q}(x_2), c)$ simply trade positions. Hence $M(\sigma J) = M(J)$ for every pooled metric M, and the group G generated by all such transpositions leaves M invariant.

The system means are not invariant. Applying σ changes

$$\hat{Q}(s_1) - \hat{Q}(s_2) \;\longmapsto\; \hat{Q}(s_1) - \hat{Q}(s_2) - \tfrac{2}{n}\big(\hat{q}(x_1) - \hat{q}(x_2)\big).$$

Choose x_1 to carry a high score and x_2 a low one, so each transposition strictly decreases the gap by $\frac{2}{n}\Delta$ with $\Delta > 0$ the within-label score spread realised by that swap. Applying $t \leq m$ disjoint such transpositions gives

$$\hat{Q}(s_1) - \hat{Q}(s_2) \;=\; \big(Q(s_1) - Q(s_2)\big) + \big(\delta(s_1) - \delta(s_2)\big) - \tfrac{2t}{n}\bar{\Delta},$$

where $\bar{\Delta}$ is the mean spread over the chosen swaps. Whenever $\frac{2m}{n}\bar{\Delta}$ exceeds the initial gap, some $t \leq m$ drives the difference strictly negative, so the judge $J_2 = \sigma_t \cdots \sigma_1 J$ orders the pair wrongly while $J_1 = J$ orders it correctly. Both have identical M. Restricting the population to $\{s_1, s_2\}$ gives $\mathrm{SRA}(J_1) = 1$ and $\mathrm{SRA}(J_2) = 0$. $\square$

Two remarks. First, the construction needs the judge to be non-constant within the label class; a judge that assigns one value to every gold-c item is unaffected, but such a judge also carries no information beyond the label. Second, the theorem is about what a pooled metric *determines*, not about what is typical: it says the map from pooled metric to ranking fidelity is not a function, so no refinement of the metric within the pooled frame can make it one.

B Stimulus-integrity detail

Full protocol of Ballı [2026], run before any statistic was computed. Round 1 is the initial build; round 2 is after the fix described in Section 3.1.

Table 3: Stimulus-integrity checks. "Verbatim copies" counts answers that normalise to the joined claim list, i.e. that were not rewritten at all.

	mean words		verbatim copies / 425		claim
Condition	round 1	round 2	round 1	round 2	fidelity
plain	77	99	325	**0**	0.995
polished	130	130	0	0	0.998
padded	286	286	0	0	0.995

Mechanical no-op check: $0/510$ corrupted claims string-identical to their true counterpart, under exact and whitespace-normalised comparison. This is the cheap oracle whose absence Ballı [2026] identify as the structural weakness of LLM-generated-negative corpora; because our corruption step is a claim substitution, it costs nothing here. Fragments under three words: 0 in all 15 conditions.

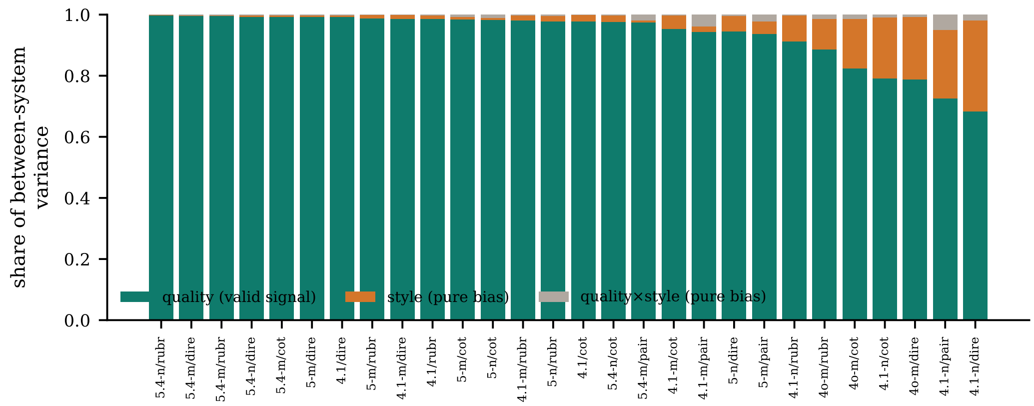

Figure 5: Two-way ANOVA on each judge's system means. Because restyling preserves quality by construction, the style and interaction components are pure bias and the quality component is the only valid signal. **Takeaway:** at the coarse quality gaps of the original grid, valid signal dominates for every judge—which is why ranking looks solved there, and why the interesting question is what happens as the gap shrinks.

Rendering-level no-ops (an answer at $k > 0$ in which the checker detects no false claim): $5/1020$ (0.5%), reported rather than removed.

We also read 20 raw items per condition by hand. One apparent fault turned out to be our own misreading and is worth recording, since it illustrates why the manual pass is not optional in either direction. In item `it0072` the corrupted claim "The Flyer used a 12-horsepower gasoline engine built by the Wright brothers" looked wrong, because $\sim$12 horsepower is historically correct. Inspecting the pair showed the horsepower is held constant across both versions and the corruption is the *builder* (Charlie Taylor $\rightarrow$ "the Wright brothers"): a correctly constructed minimal entity substitution. From an aggregate view this is invisible; from the raw pair it takes ten seconds.

C Error decomposition

D Restricted transformation family

A reader may hold that verbosity genuinely degrades an answer, so that our `padded` transform is not quality-preserving and part of what we call bias is signal. We therefore repeat the analysis with $\mathcal{T}' = \{\texttt{plain}, \texttt{polished}\}$, which are close in length (99 vs 130 words) and differ essentially in formatting.

Table 4: Restricted family $\mathcal{T}'$ versus the full family $\mathcal{T}$.

	$\mathcal{T}$ (3 styles)	$\mathcal{T}'$ (2 styles)
Mean quality share of between-system variance	0.935	0.978
Minimum quality share over judges	0.684	0.711
Spread of resolution limits across judges	60×	35×
$\rho(\delta_{\text{style}},\ \text{limit})$ within family	0.936	0.786
R^2 predicting the limit within family	0.927	0.876
Residual s.d. (% of factual content)	3.3	2.5

The phenomena survive: even with the disputed condition removed, resolution limits still span $35\times$ and the label-free estimator still recovers them far more precisely than any gold-label predictor does (2.5% residual against $\sim$6.3%; Table 1).

One cross-family result is worth stating because it could be mistaken for a failure. Using the restricted estimator $\delta^{\mathcal{T}'}_{\text{style}}$ to predict the *full-family* limit gives only $\rho = 0.489$. This is the theory behaving as advertised rather than breaking: Theorem 5 is sound relative to the declared family, and a family that contains no verbosity contrast cannot bound a confound driven by verbosity. The estimator predicts exactly the confounds it is asked about. The practical consequence is that $\mathcal{T}$ must be declared, and declaring it is precisely the act of committing to a quality construct.